

Review article

Anti-UAV detection and identification technology: Fundamentals, methods and challenges

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ABSTRACT

The rapid development of unmanned aerial vehicle (UAV) technology has driven its widespread application in military, commercial, and civilian domains, such as surveillance, logistics, and disaster response. However, this growing popularity has also raised serious security and privacy concerns, as UAVs can be misused for smuggling, espionage, terrorism, and other illegal activities. As a result, anti-UAV detection and identification technologies have become a critical research area for monitoring, identifying, and mitigating the threats posed by rogue drones. This paper provides a comprehensive review of the latest advancements in anti-UAV detection and identification technologies. To deepen the understanding of these technologies, the paper first introduces the UAV system (UAS) framework and its corresponding anti-UAV systems. Subsequently, it presents an in-depth analysis of the primary detection and identification methods, including those based on image processing, radio frequency (RF) signals, radar, infrared characteristics, acoustic features, and multi-modal fusion. The advantages, limitations, and application scenarios of each method are systematically compared and summarized. Furthermore, this paper identifies the major challenges faced by current anti-UAV detection technologies, such as detection accuracy, range, robustness, and cost, and explores potential directions for future research and development. By synthesizing these insights, this paper aims to provide researchers and practitioners with a comprehensive understanding of the current state of the art in anti-UAV technologies, serving as a valuable reference to advance research progress and foster innovation in this critical field.

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1. Introduction

The unmanned aerial vehicle (UAV) market has evolved quickly due to the technology's rapid growth. Numerous industries, including tracking, aerial photography, agriculture and forestry operations, military operations, traffic monitoring, rescue and disaster relief, emergency communication, positioning services, aerial surveying and mapping, and security supply [1], use it extensively due to its small size, light weight, good concealability, and strong autonomous flight ability [2]. Fig. 1 visually represents various UAV applications, including surveillance, logistics, and disaster response, highlighting their versatility across industries.

But as the UAV industry grows rapidly, there is a greater chance of an unintentional collision with an object, which raises a number of safety concerns. Some UAVs are used for smuggling, navigational interference, espionage, invasions of privacy, terrorist attacks, and other activities that pose a threat to national security [3,4]. Moreover, the deployment of UAVs may inadvertently or intentionally violate some national security rules [5]. "Illegal flights" refer to flights conducted by individuals who have not obtained a private pilot license or by aircraft that are not legally registered. Fig. 2 illustrates the four main safety risks associated with drones operating in "illegal flight".

Attackers may attach bombs or other dangerous materials to drones to carry out attacks. In addition, some criminal also use drones to smuggle illegal materials. Operators may also use drones with high-definition cameras to spy on other people's privacy [6]. Some UAVs may violate international human rights conventions when performing military missions and are called malicious UAVs [7]. Some of the threats posed by drones are shown in Fig. 3.

In order to effectively deal with the challenges brought by the development of UAVs, it is necessary to further develop the detection and countermeasures technology of UAVs. Anti-UAV technology includes UAV identification technology, UAV positioning technology and UAV interception technology. UAV recognition is to collect data through sensors to quickly identify whether there is drone intrusion in the relevant area. UAV positioning refers to the dynamic acquisition of the space coordinates of the invading UAV, and then the implementation of interference or interception. UAV interception is the use of optical, electromagnetic and other measures to achieve the intrusion of UAV interference, interception, drive away and even destroy. Some reviews and surveys on UAV detection have been published in recent years, and the differences between this paper and them are shown in Table 1.

In addition, in the field of UAV detection and tracking, the diversity and richness of datasets are crucial for the development and performance evaluation of algorithms. In order to promote the research progress in this field, several research teams and organizations have constructed a series of representative datasets, which not only cover different environmental conditions and mission requirements, but also provide rich annotation information and diverse data types, providing a solid foundation for the development and evaluation of UAV detection and tracking algorithm development and evaluation provides a solid foundation. Table 2 summarizes several existing datasets for anti-UAV detection and tracking, including their features, data size, data types, and application scenarios. These datasets play an important role in drone detection, classification, tracking, and trajectory estimation, and provide valuable resources for researchers. By comparing these datasets, we can better understand the current state of research and future directions in the field.

However, the existing anti-UAV detection technology has some limitations in accuracy, range, cost, robustness and reliability, so we investigate the anti-UAV detection methods. The contributions of this work are as follows.

(1) Overview of technological advances and applications: A comprehensive overview of existing anti-drone detection methods, in which include image processing, RF signals, radar techniques, infrared signatures, and acoustics, and gives their advantages and comparative disadvantages.

(2) Technological progress and application: This paper reviews the latest progress and related applications of UAV detection and identification technology, and provides researchers with a comprehensive understanding of this field.

(3) Introduction to the framework of UAV and anti-UAV systems: Firstly, the system framework of UAV and anti-UAV is introduced to help readers better understand the application scenarios of various detection technologies.

(4) Discussion on Challenges and Future Development Prospects: The challenges faced by the existing UAV detection and identification technologies are discussed, and some suggestions are put forward for the future research direction of anti-UAV detection and identification systems.

The structure of the review is shown in Fig. 4. In addition, the abbreviations mentioned in the article are listed in Table 3 for ease of reference.

The following specialized terms are newly included for further clarification:

(1) Gyroscope Resonance: Vibrations occurring at the natural frequency of a gyroscope, which may lead to functional instability in UAVs. This phenomenon can be exploited in counter-UAV systems to disrupt flight stability through acoustic or mechanical interference.

(2) RF Detection: The use of radio frequency signals to detect and identify UAVs by analyzing their communication patterns or signals emitted during operation. RF detection is particularly effective for UAVs that rely on remote control or telemetry links.

(3) Soft Kill: Non-physical countermeasures aimed at disabling or impairing UAV functionality without causing physical damage. Examples include jamming, electromagnetic interference, and spoofing, which disrupt UAV communication or navigation systems.

(4) Hard Kill (Physical Destruction Techniques): Refers to counter-UAV methods that physically destroy or disable the target using kinetic weapons such as missiles, projectiles, or lasers. Hard kill methods are highly effective in ensuring complete neutralization of UAV threats but often entail high operational costs and the risk of collateral damage.

2. Challenges

To safeguard people, assets, and critical infrastructures from potential threats posed by the unauthorized use of drones, it is crucial to develop efficient and reliable drone detection methods. The benefits and limitations of various detection techniques are summarized in Table 4, providing a comprehensive comparison of their capabilities.

The existing drone detection methods mainly face the following challenges:

2.1. Environmental challenges

1. **Low Visibility Conditions:** Image-based drone detection methods often suffer from performance degradation under poor visibility conditions, such as rain, haze, or nighttime.

Table 1
Main contributions of Recent Surveys/Reviews.

Ref.	Year	Journal	Contributions
[8]	2018	IET Radar, Sonar and Navigation	Review of radar classification and RCS for small UAVs.
[9]	2021	IEEE Access	Overview of UAV detection and recognition techniques.
[10]	2021	Ad Hoc Networks	Analysis of UAV attack methods and defense techniques, including radar, RF, video, and audio surveillance.
[11]	2022	IEEE Sensors Journal	Summary of UAV detection technologies, excluding infrared.
[12]	2023	Shanlax International Journal	Exploration of UAV applications, challenges, and swarm-based solutions.
[13]	2024	Computer Science Review	Review of unethical UAV behaviors and deep learning-based detection methods.
[14]	2024	Remote Sensing	Comprehensive analysis of ML/DL-based UAV detection, including datasets, challenges, and future directions.
This review	2025	Under review	Systematic framework for UAVs and counter-UAVs, covering detection methods and multimodal fusion.

Table 2
Summary of UAV-related Datasets.

Dataset	Ref.	Characteristics	Data scale	Data types	Application scenarios
Anti-UAV	[15]	Multi-modal dataset with synchronized RGB and IR video pairs	318 video pairs, 580K annotations	RGB, IR	UAV tracking and detection
Drone-vs-Bird	[16]	Video dataset for UAV and bird distinction	5 training, 1 test video	RGB	UAV and bird classification
DUT Anti-UAV	[17]	Visible light dataset for UAV detection and tracking	10K images, 20 sequences	RGB images, videos	UAV detection and tracking
MMAUD	[18]	Multi-modal dataset integrating vision, LiDAR, radar, and audio	50 sequences, 1700 s	RGB, LiDAR, radar, audio	UAV detection, classification, and trajectory analysis
MAV-VID	[19]	UAV video dataset for detection and tracking	64 videos, 40K frames	RGB	UAV detection and tracking
AntiUAV600	[20]	Large-scale dataset with thermal infrared video sequences	600 sequences, 723K frames	Thermal infrared	UAV perception in complex environments

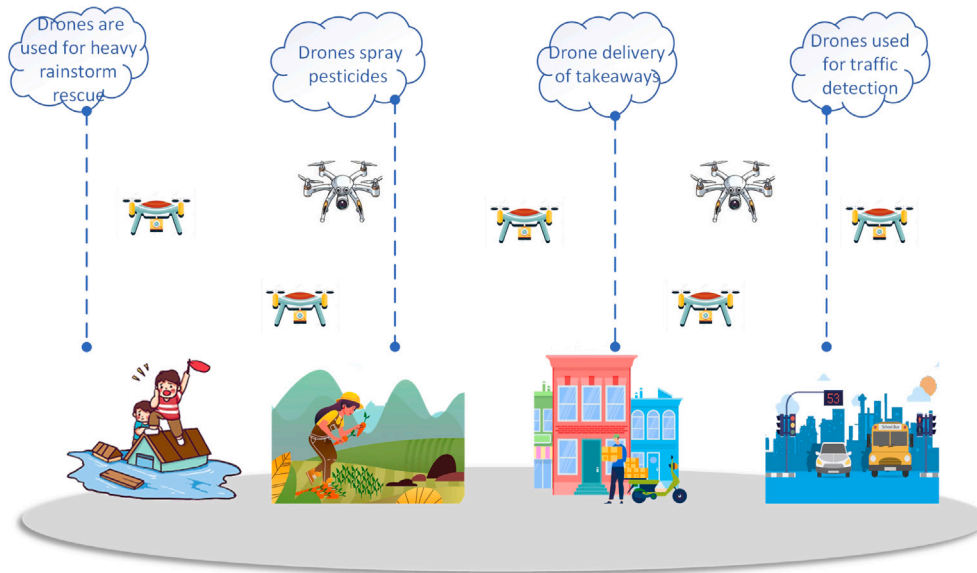


Fig. 1. A portion of the scenarios for using UAVs.

- Lighting and Weather Conditions:** Lighting and weather conditions can blur or obscure the visual features of the target object.
- Target Separation in Complex Backgrounds:** In complex environments (e.g., buildings, trees, or birds), the features of the background may overlap with those of drones, leading to increased false positive rates and reduced system reliability.
- Small Target Detection:** Due to the small size and lightweight nature of drones, their radar cross-section (RCS) is much smaller than larger objects like airplanes, making them harder to detect. Additionally, stealth technologies like radar-absorbing materials

can further reduce radar signatures, allowing drones to evade detection.

2.2. Technical challenges

- Sensor Limitations:** Radar systems face interference from other equipment in complex environments, while power transmission regulations further restrict their sensitivity and range. Acoustic sensors, on the other hand, struggle in noisy environments or when drones fly at low speeds or hover silently.
- Multimodal Synchronization:** Multimodal fusion faces difficulties due to differences in sampling frequency, time delays, and

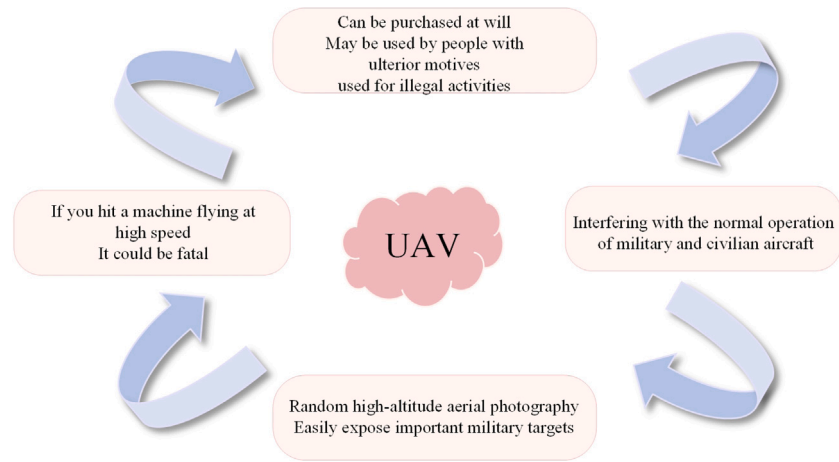


Fig. 2. Four safety risks associated with UAV “illegal flight”.

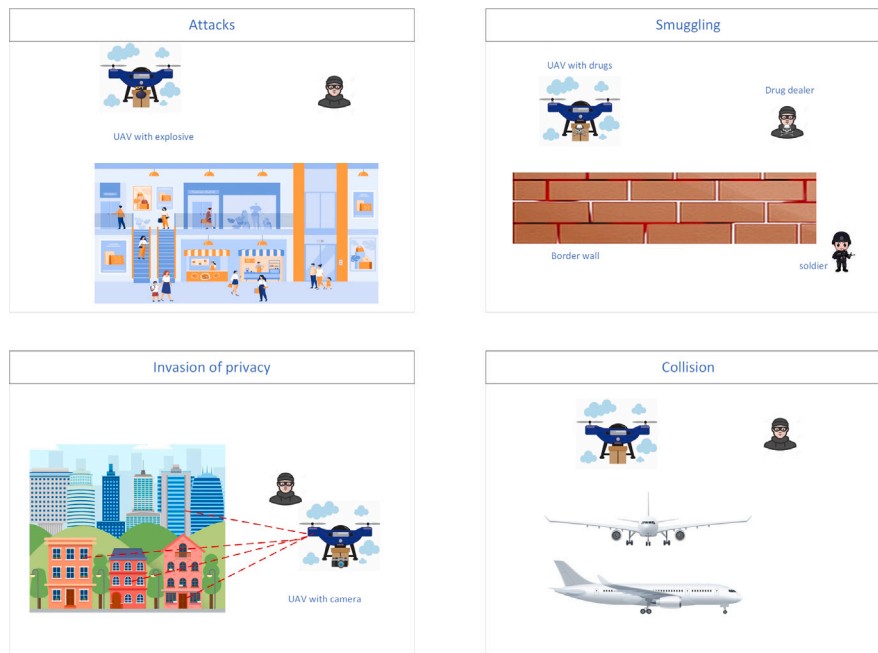


Fig. 3. Some of the threats posed by drones: (a) Drone terrorist attacks (b) Drones transport drugs; (c) Drones violate personal privacy (d) Drone interfering with aircraft navigation.

spatial perspectives between sensors. These discrepancies make direct data fusion challenging without effective synchronization mechanisms.

3. **Limited Dataset Diversity:** Existing datasets are often small and lack diversity, making them insufficient to represent UAV characteristics in different scenarios (e.g., various drone types, flight altitudes, and speeds).
4. **Impact of Distance:** Detection accuracy often decreases significantly as the distance between the UAV and sensor increases, resulting in a limited detection range.

3. UAV system framework

The framework of a traditional UAV system usually consists of two parts: (1) the UAV and (2) the ground controller [8], as shown in Fig. 5. Typically, the UAV is equipped with a power supply, an on-board controller, several sensors (including inertial sensors, magnetometers, gyroscopes, and GNSS), and a wireless transceiver, which receives control signals from the wireless transmitter through a receiver and manipulates the rotor's speed according to the user's control using the

flight controller, which ensures smooth flight and safe landing of the UAV, complete path tracking and target detection [21].

The remote control can be manually operated, or it can be a semi-automatic or fully automatic control unit composed of a control unit and a wireless transceiver. Therefore, drones can be controlled remotely through wireless technologies such as mobile phones (Wi-Fi, cellular 3G, 4G or 5G), Bluetooth, radio channel frequencies, or the Internet of Things [22]. Therefore, the RF signal transmitted through the communication link can also be used for drone detection.

According to differences in performance, size, weight, flying height, mission radius, and the purpose of usage, unmanned aerial vehicles (UAVs) can be classified into several categories. Unmanned aerial vehicles (UAVs) now in use can be divided into five primary categories based on their size and quality: micro, light, small, medium, and large UAVs, as is shown in Table 5. Different types of unmanned aerial vehicles (UAVs) exhibit significant differences in performance, size, flight altitude, and mission radius. These differences directly affect the difficulty of UAV detection. For instance, micro UAVs are challenging to detect using radar due to their small size and low flight altitude, whereas large UAVs, operating at higher altitudes, are more likely to

Table 3

List of Abbreviations.

Abbreviation	Full form
UAV	Unmanned Aerial Vehicle
SVD	Singular Value Decomposition
SNR	Signal-to-Noise Ratio
MAC	Media Access Control
RF	Radio Frequency
FMCW	Frequency-Modulated Continuous Wave
RCS	Radar Cross-Section
MFCC	Mel-Frequency Cepstral Coefficients
LPCC	Linear Predictive Cepstral Coefficients
DCT	Discrete Cosine Transform
FFT	Fast Fourier Transform
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
CRNN	Convolutional Recurrent Neural Network
ICA	Independent Component Analysis
PSD	Power Spectral Density
RMS	Root Mean Square
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
IVA	Independent Vector Analysis
SLL	Sound Source Localization
MUSIC	Multiple Signal Classification
DOA	Direction of Arrival
ECG	Electrocardiogram
IoU	Intersection over Union
YOLO	You Only Look Once

be detected through infrared signals. Therefore, detection and identification technologies must be adapted to specific UAV types, employing appropriate sensors and algorithms to achieve efficient target acquisition. Based on this, we will subsequently explore several major UAV detection and identification technologies in detail, along with their applicability and limitations in addressing different types of UAVs.

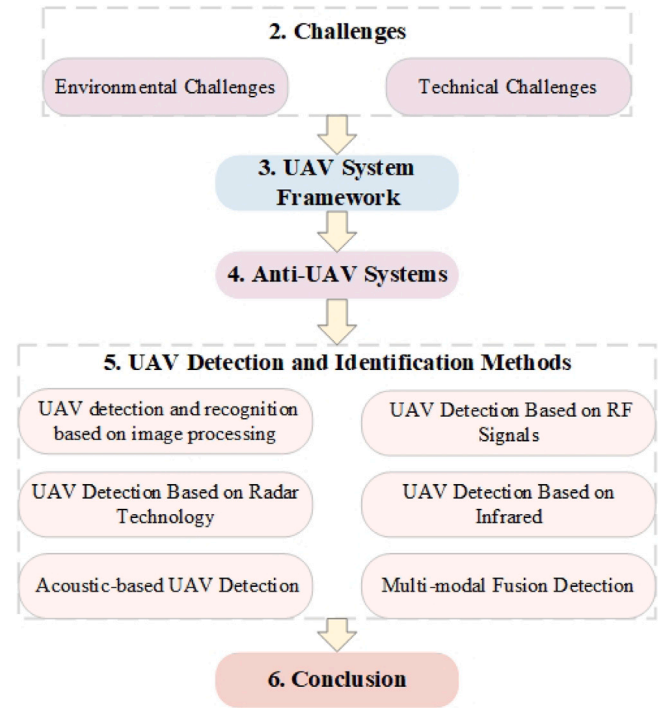
4. Anti-UAV systems

As is shown in Fig. 6, hard kill and soft kill are the two categories of technical approaches into which UAV countermeasures can be separated [23]. Hard killing techniques center on protecting military systems and concentrate on physical kinetic energy attacks, such as the direct destruction of UAVs by anti-aircraft guns, anti-aircraft missiles, and other weapons to achieve hard physical damage [24]. However, soft kill techniques concentrate on using the light, heat, sound, electricity, and other characteristics of the unmanned aerial vehicle (UAV) itself. They also develop unique technologies for identification and attack to carry out non-physical interference or control of the UAV, with the goal of weakening or destroying its function [25]. These techniques are primarily used to protect civilian fields or critical infrastructure [26].

The anti-UAV system mainly includes three types: monitoring and control, direct destruction and interference blocking [27]. Among them, the monitoring and control class controls and guides the safe return of the UAV by intercepting and blocking the transmission code used by the UAV, thus effectively avoiding the crash risk of the UAV [28]. The direct destruction category mainly relies on missiles, laser weapons, microwave weapons, combat drones, conventional fire and other means to directly destroy drones. Interference blocking refers to cutting off the communication between the UAV and the remote control platform by transmitting directional high-power interference radio frequency, forcing the UAV to land or return by itself [9] (see Fig. 7).

4.1. Monitoring and control

During the automatic return of the UAV, the UAV needs to communicate with the ground control station in real time in order to receive

**Fig. 4.** The structure of the review.

return commands as well as to send status reports. During this process, if there is a loss of communication signals, the UAV will not be able to receive further instructions, which may result in failure to return safely on the intended path [29]. Therefore, monitoring and control type technologies can be useful in such situations to prevent the UAV from continuing along its original flight path by intercepting and blocking the transmission code used by the UAV, thus avoiding potential danger.

4.2. Interference blocking class

At present, the interference blocking class mainly includes electromagnetic interference, navigation signal interference, acoustic interference and other types of interference. Among them, electromagnetic interference, as an efficient means, can accurately block the communication link between the UAV and the control station, including key information such as the remote control signal, data transmission and image transmission [30], forcing the UAV to automatically switch to the protection mode after losing the signal, so as to realize the safe forced landing or effective driving of the UAV. According to the actual demand, electromagnetic interference can be implemented in a directional or omnidirectional interference mode [31], which can not only effectively deal with a single UAV within the operating distance, but also have the ability to efficiently strike the “UAV cluster”.

A civil UAV usually uses satellite navigation and positioning systems to determine its position. By implementing navigation signal jamming, we can effectively block the communication between the UAV and navigation system. When the UAV loses the navigation signal, it will not be able to accurately locate, which will directly affect its flight control system, thus limiting the flight capability and range of the UAV [32]. Due to the weak strength of the navigation signal itself, this jamming method is relatively easy to achieve. However, it is worth noting that this interference mode may also have adverse effects on the navigation terminals of other devices in the range of action, so extra caution is required when using it. The gyroscope is installed inside the drone, which is responsible for helping the drone accurately sense and judge its flight state. The UAV relies on the feedback information

Table 4
Detection methods for UAVs: Advantages and Disadvantages.

Detection method	Advantages	Disadvantages	Detection range
Image feature	High recognition accuracy, reliable performance	Weather-dependent, can be blocked	<3 km
RF feature	Not affected by obstacles	Cannot detect autonomous drones, high cost	3–50 km
Radar feature	Weather-independent, long range	Occlusion issues, equipment limitations	1–20 km
Infrared features	Weather-independent, low cost	Low recognition accuracy	<1 km
Acoustic features	Low cost, miniaturized, combinable with other methods	Short range, noise-sensitive	<0.2 km

Table 5
Classification of Unmanned Aircraft Systems.

Type	Parameters		
	Take-off mass (kg)	Flight height (m)	Mission radius (km)
Micro drone	< 2	< 100	5
Mini drone	2–20	< 1000	25
Small drones	20–150	< 1500	50
Tactical drones	150–600	< 5000	200
Medium altitude long endurance drones	> 600	< 15000	> 500
High altitude long endurance drones	> 600	< 20000	Worldwide

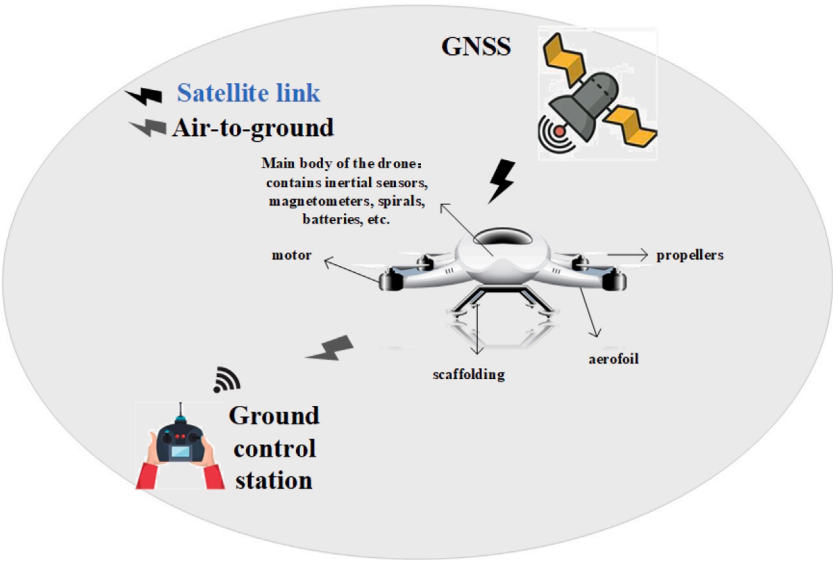


Fig. 5. Classic structure of unmanned aerial system.

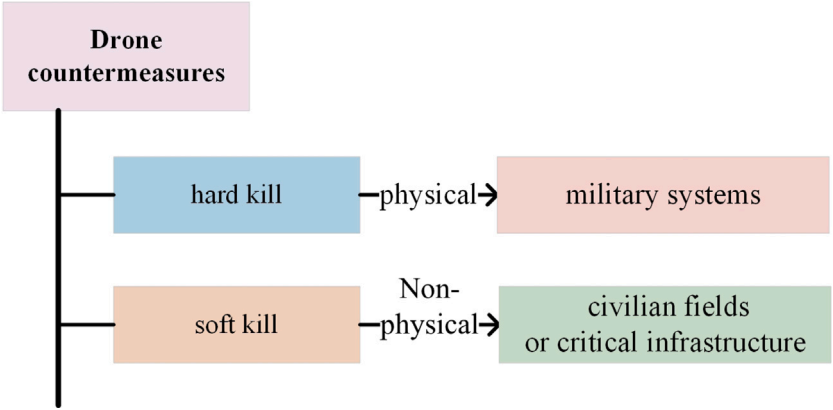


Fig. 6. Classification of anti-drone systems.

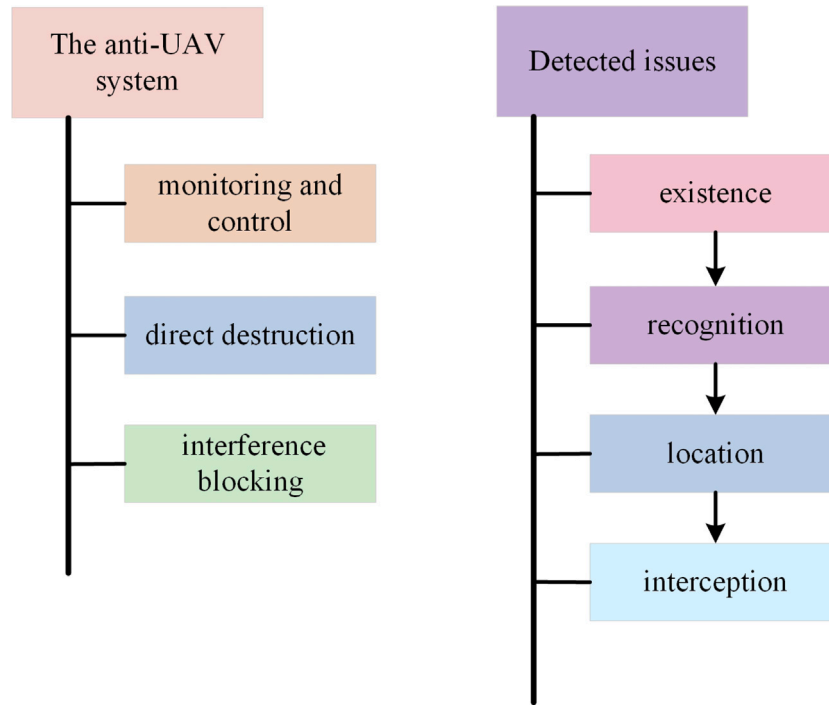


Fig. 7. Anti-UAV system and detected issues.

provided by the gyroscope and the control system to maintain the balance and stability of its flight [33]. Based on this principle, if the normal operation of the gyroscope can be successfully interfered with or destroyed, the UAV will lose its normal flight ability [34].

Acoustic interference, as a potential interference mode, is to emit sound waves matching the operating frequency of the gyroscope, causing the resonance effect of the gyroscope, and then making it lose its normal function. This interference mode will cause the gyroscope on the drone to fail to work properly, thus putting the drone at risk of crashing. However, at present, the cost of this jamming technology is high, so it is not suitable for large-scale or widespread promotion and use.

4.3. Destroy directly

Various means, such as missiles, laser weapons, microwave weapons, combat drones and conventional fire, can effectively carry out targeted attacks on drones and directly destroy them [35]. However, these means require extremely high targeting accuracy when implemented and are relatively costly. More critically, the fall of a drone may trigger collateral damage or a chain reaction [36], which is unacceptable in the civilian sector. Therefore, while these methods have a wide range of applications in the military field, in the civilian field, the direct destruction of drones is hardly feasible.

During the flight of the UAV, the motor operation will generate heat. The rapid rotation of the UAV rotor will produce noise. The UAV communicates with the remote control terminal through the radio frequency signal, and these processes will produce different characteristic information, which can be collected by different sensors, and then realize the detection and identification of the UAV. The development history of UAV detection is shown in Fig. 8.

5. UAV detection and identification methods

According to the different types of sensors in the UAV detection system, the existing UAV detection and recognition technologies can be divided into the following categories: UAV detection and recognition based on image processing, UAV detection and recognition based on

radio frequency signal processing, UAV detection and recognition based on radar technology, UAV detection and recognition based on infrared characteristics, and UAV detection and recognition based on acoustics, which are shown in Fig. 9.

5.1. UAV detection and recognition based on image processing

During the flight of the UAV, internal hardware such as the motor and battery will radiate a large amount of heat, and thermal imaging can confirm the existence of the UAV by detecting this heat [37]. UAV detection schemes based on image processing make use of computer vision and deep learning to detect UAVs. Daylight cameras and infrared or thermal imaging technologies are usually used to capture UAV images or videos in the monitored area according to the object's appearance features [38], that is, color, shape, contour, etc., or its motion features across continuous frames [39]. The object detection technology based on computer vision is used to detect and track the UAV. The research performance of image-based UAV detection and recognition is shown in Table 6.

Sapkota et al. proposed a method of online identification of small UAVs using AdaBoost combined with an online vision tracking algorithm, sensor fusion and state estimation method for rapid UAV detection [40]. This method provides real-time monitoring capabilities but lacks detailed validation on complex backgrounds or dynamic environments. Adrian Carrio proposed a new UAV detection method that relies only on depth maps and can provide 3D positioning of drones, which can be used to implement target detection, and the model can generalize different types of drones well. In addition, the author also discloses a 6K synthesized depth map dataset using three different UAV models for further exploration of learning in this field [41]. BH Sheu et al. proposed a two-axis structure of UAV automatic tracking, which utilizes visual image processing techniques to track and lock on to the dynamic displacements of UAVs [42]. This system shows promise but is significantly affected by weather conditions and lacks validation on complex backgrounds. Xue Shan et al. proposed a convolutional neural network-based image recognition method for UAV small samples, and conducted comparative experiments using convolutional neural networks and support vector machines respectively, and found



Fig. 8. The evolutionary history of drone detection technology.

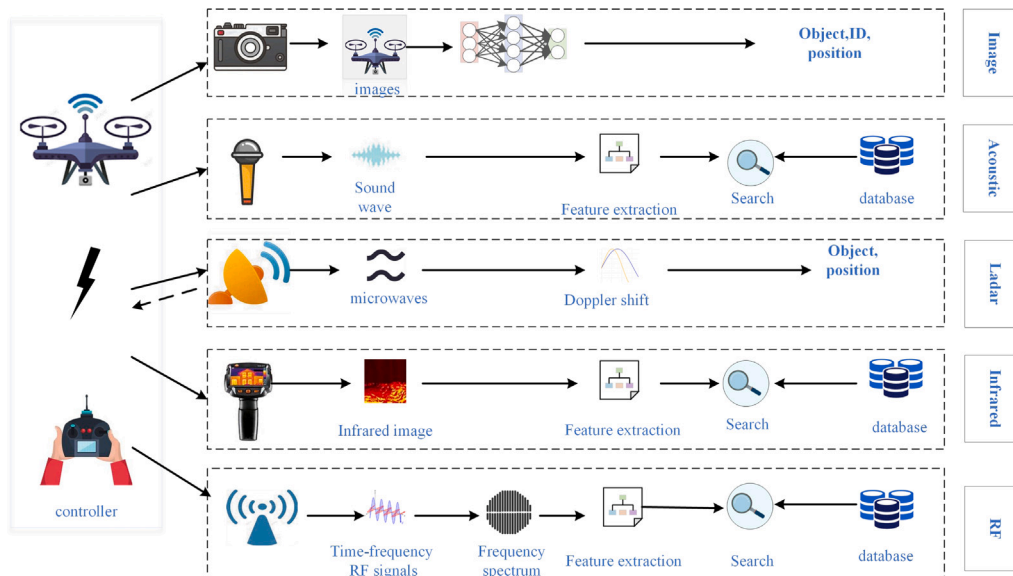


Fig. 9. Typical UAV detection and identification technology.

that the convolutional neural networks used performed well [43]. However, the method's performance on small target detection remains unclear. S. Srigrarom et al. proposed a real-time multi-camera system that can detect, track and locate UAVs in 3D space with an average position error rate of only 8% [44]. Farhad Samadzadegan proposed a method based on deep learning in order to the effective detection and identification of drones and birds. YOLO V4 was used to detect and identify targets, which solved the problem of drone detection under background congestion and area hiding conditions and the challenge of confusion between drones and birds in visible images [46]. To solve the problem of tiny UAV detection, Muhammad Waseem Ashraf et al. proposed a novel two-stage segmentation method, which uses spatio-temporal attention cues to realize UAV detection [47]. This method provides a new approach for detecting small UAVs but lacks validation on complex backgrounds or dynamic environments. Tushar Sangam et al. proposed a simple and efficient end-to-end optional UAV

video detection framework, TransVisDrone. This framework uses the CSPDarkNet-53 network to learn target-related spatial features and improves detection performance with the VideoSwin model, which learns the spatiotemporal dependencies of UAV motion. The method achieves good performance on three publicly available datasets [48]. To address the issue of interference in UAV detection caused by aircraft, birds, and other similar airborne objects, Burchan Aydin et al. proposed a UAV detection method based on YOLOv5, which leverages data augmentation and transfer learning techniques to enhance detection accuracy and speed [49]. This method addresses the issue of interference from similar airborne objects but lacks validation on small targets or complex backgrounds.

The UAV detection and recognition technology based on image processing offers advantages such as low system cost, high recognition accuracy, reliable performance, a wide monitoring range, and a high level of technological maturity. However, its detection performance is

Table 6

Comparison and summary of UAV detection and recognition technology based on image processing.

Ref.	Detection target	Database	Method	Performance
[40]	Drone detection and tracking	Custom dataset	Adaboost	Real-time monitoring
[41]	UAV detection	Custom dataset	YOLOv2	Accuracy: 98.7%, Recall: 74.7%, IoU: 86.41%
[42]	Drone tracking	Custom dataset	System	Weather affected
[43]	Drone identification	Custom dataset	CNN	Accuracy: 95.9%
[44]	Drone detection, tracking, and positioning	Custom dataset	YOLOv3	Accuracy: 92%
[45]	UAV detection and positioning tracking	Custom dataset	Improved YOLOv3	Accuracy: 83.24%, Recall: 88.15%
[46]	UAV detection and identification	Custom dataset	CNN	Accuracy: 83%, mAP: 84%, IoU: 81%
[47]	UAV detection	NPS-Drones and FL-Drones	ResNet50	Precision: 0.92/0.84, Recall: 0.91/0.76, F1: 0.92/0.80, IoU: 86.41%
[48]	UAV detection	NPS, FL-Drones, and AOT datasets	CSPDarkNet-53 and VideoSwin	mAP: 0.95/0.75/0.80, IoU: 86.41%
[49]	UAV detection	Custom dataset (2395 images)	YOLOv5	Accuracy: 91.8%, Recall: 87.3%, mAP: 90.4%, IoU: 86.41%

Table 7

Comparison and summary of UAV detection and recognition technology based on RF signals.

Ref.	Detection target	Method	Performance
[53]	Drone detection	DL	Accuracy: 97.5%
[54]	UAV detection	LTS, AGC, SVD, SA, SFA	Accuracy: 2400 m – 100%, 3000 m – 90%
[55]	Drone identification and classification	YOLO	Detection: YOLO > GoF, Classification: YOLO ≈ DRNN
[56]	Drone detection	ML	Height > 60 m: Accuracy = 100%
[57]	UAV detection and identification	DNN	Accuracy: 99.7%
[58]	UAV detection	CNN	Detection Accuracy: 99.85%, Classification: 98.53%, Mode Identification: 95.33%
[59]	UAV detection and identification	Integrated System	GNB: 99.6%, SVM: 99.2%, NN: 98.8%

significantly affected by external factors, particularly under adverse weather conditions (e.g., fog, rain) or in low-visibility environments, where the method faces considerable challenges. Furthermore, the presence of objects such as birds or kites may lead to false detections or confusion in the recognition process [6].

5.2. UAV detection and identification based on RF signal

UAVs typically use radio frequency signals from 2.4 GHz to 5 GHz [50] to communicate with the controller. Wi-Fi-equipped drones use 5.4 GHz radio frequency signals, and 5G drones use 3.5 GHz radio frequency signals. Some less common frequency bands are in the 1.2 GHz and 1.3 GHz ranges [51]. By extracting spectral features, the statistical distribution of RF signals is completed, which is used as the RF features of the UAV [52]. RF sensors can be used to monitor the signals transmitted between the UAV and its controller. The anti-UAV detection and identification framework based on RF signal is shown in Fig. 10.

There are two ways to scan an RF signal: the first involves listening for and decoding data packets sent back and forth between the drone and its controller. More specific information, including the identification, location, and speed of the drone as well as the data it contains, such a video stream, may be given because the complete packet has been decoded. Nonetheless, several nations forbid using this technique. The other is that the RF receiver merely records the RF signal and looks for a certain pattern in the recorded data; it does not decode the packet. The information that this approach can provide is restricted to identifying the presence of a drone, estimating its location, and predicting the drone's brand and model. Due to regulatory constraints, the second method is mostly used for drone detection. The performance of UAV detection and recognition based on RF signals is shown in Table 7.

P Nguyen et al. [52] studied a system for autonomous detection of UAVs using radio-frequency wireless signals, and proposed two solutions. One method was active tracking, in which the system monitored the reflected components after sending radio signals; Another method is to receive, extract and analyze the observed radio signals through passive monitoring. Su Zhigang et al. proposed an innovative RF signal detection model to solve the problem of low RF signal detection accuracy of UAVs under low SNR conditions, which combined a multi-branch convolutional structure and attention mechanism to significantly improve the RF signal detection accuracy of UAV under low SNR conditions [53]. Shengying Yang et al. proposed an improved RF based UAV detection method to eliminate interference by background filtering, and then adopted SVD (singular value decomposition) and average filtering to reduce noise and improve signal-to-noise ratio (SNR) [54]. Sanjoy Basak et al. developed a radio frequency (RF) dataset encompassing commercial drones and WiFi signals, utilizing the YOLO-lite architecture to detect and classify UAV transmission signals' frequency features. By transforming RF signals into spectrograms for YOLO model training, YOLO outperformed Goodness-of-Fit (GoF) spectrum sensing in multi-signal detection and matched the classification accuracy of Deep Residual Neural Networks (DRNN). This integration of signal processing and deep learning presents an innovative approach to drone RF signal detection [55]. Henrik Ryd'en applied two machine learning models, logistic regression and decision tree, to realize UAV recognition according to the features measured by radio, but in the low-altitude field, the detection accuracy was low [56]. Syed Samiul Ala et al. [57] proposed an end-to-end deep learning model for UAV detection and identification based on the RF characteristics of UAV, and used multi-scale feature extraction technology without manual intervention to extract features, so that the model could obtain good signal generalization and decision-making ability in a short computing time. It achieves effective detection and recognition of UAV signals in real-time scenarios. Sanjoy Basak [58] et al. Thien Huynh-The et al. proposes a

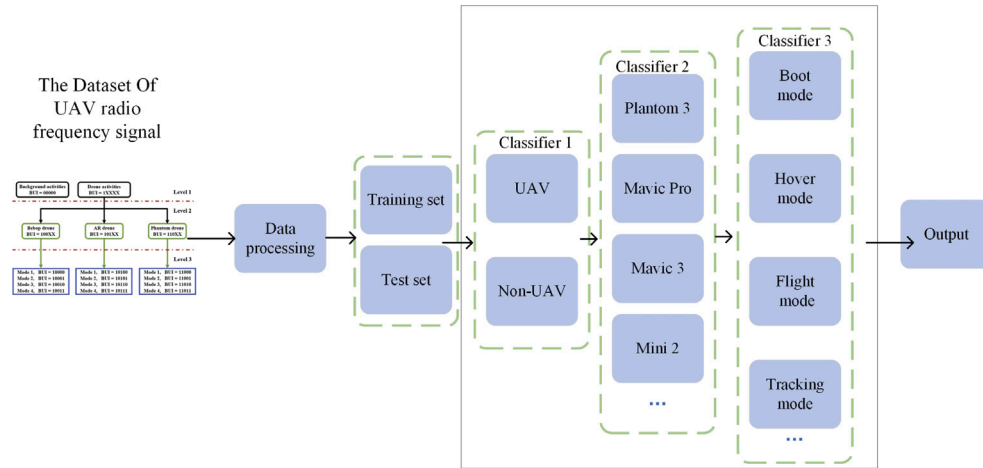


Fig. 10. Anti-UAV detection and recognition framework based on RF signal.

new CNN architecture named RF-UAVNet, which has a new structure of multi-level pooling just like multi-level jump connection, and realizes 10x cross-validation experiment on the data set, and the system performs well. Phuc Nguyen et al. Yu Ningning et al. [60] disclosed a UAV radio frequency signal dataset, which was used to record the real data of UAV multi-frequency communication for detecting low-altitude UAV. In Ref. [59], Yash Vasant Ahirrao et al. proposed a real-time signal processing system combining hardware and software to realize the detection, tracking and positioning of UAVs. The signal sensitivity reached -120 dBm and the maximum detection distance was 300 m, which was the highest level in the literature, and the system used commercially available middle and low end hardware components. Lower costs while still achieving high performance. [61] systematically investigates the Unmanned Aerial Vehicle (UAV) surveillance system based on RF (Radio Frequency) and WiFi technologies, focusing on UAV detection and tracking techniques and discusses in detail the application of RF and WiFi technologies in UAV surveillance.

It is also possible to identify the drone's Media Access Control (MAC) address, but only drones with open MAC addresses can be detected. Furthermore, it is getting harder to create and maintain a comprehensive database of drone MAC addresses due to the growing variety of drones. In order to avoid detection, the MAC addresses of some drones are deceptive [62].

RF inspection has various advantages over other methods, such as no need for LOS, longer detection range, robustness to weather conditions, and higher accuracy. However, in the actual detection process, the performance of RF detection largely depends on the interference from other RF sources and the impact of rough terrain on the RF signal of the UAV.

5.3. UAV detection and recognition based on radar technology

Radar (radio detection and ranging) is commonly used to detect aircraft in both military and civilian applications, such as aviation, and is therefore generally considered a mature tool for detecting UAVs. Radar emits radio waves and picks up reflected waves from objects. The receiver detects the presence, distance, and speed of the object by analyzing the Doppler shift caused by the moving object [63]. Radar can be divided into passive radar and active radar. Active radar transmits signals and receives reflected signals to detect targets [58], while passive radar itself does not transmit any signals and relies on other signal sources, such as analog television, FM radio, cellular signals, and digital audio and video broadcasting [51]. Active radar is commonly used for UAV detection and is generally divided into two categories: pulse radar and frequency modulation continuous wave (FMCW) radar. Pulse radars emit short pulses and measure the time

delay between the transmitted and received signals to detect the target, while FMCW radars emit continuous waves and measure the frequency difference between the transmitted and received signals to determine the distance to the target [64]. Ka-band (26.5 GHz 40 GHz), Ku-band (12 GHz 18 GHz), X-band (8 GHz 12 GHz), C-band (4 GHz 8 GHz) and S-band (2 GHz 4 GHz) are usually used. The anti-UAV detection and identification framework based on RF signal is shown in Fig. 11.

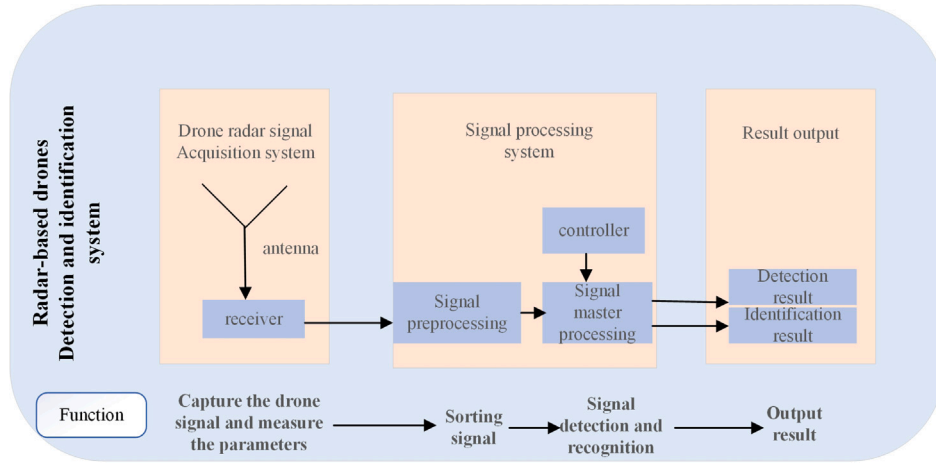
M Jian et al. used the joint range-Doppler azimuth processing technology to detect and track the micro-UAV, and used the features extracted from the range-Doppler azimuth domain for UAV recognition [65]. M et al. proposed a 35 GHz frequency-modulated continuous wave (FMCW) radar equipped with a fixed antenna, and found that this scheme could effectively detect UAVs, and the detection efficiency of the system could be improved if circular polarized antennas were used [66]. M Fennel et al. proposed a system setup based on MIMO OFDM radar, which can be used for UAV detection and general detection of unmanned aerial vehicles (UAVs) [67]. Shi Xiongwei studied the detection methods of rotary-wing UAVs in motion state and hovering state under FM-CW radar, and designed and developed the target detection software system of UAVs with LFM-CW radar [68]. Ann Janeth Garcia et al. proposed an improved radar UAV recognition system based on convolutional neural network. The most accurate classification results were obtained by adjusting the number of filters, the number of combination layers and the number of feature extraction blocks by using FM continuous wave based on the 8.75 GHz band [69]. This paper [70] proposes a deep learning technology based on CNN, which uses spectral images generated by STFT and radar signals as input, combined with image data of visual detection, to solve the detection problem of small radar cross-sectional area (RCS) targets (such as drones and birds). The literature discusses techniques for detecting, classifying, and tracking airborne threats, such as drones, using radar and communication systems, including RF signals. [71] analyzes in detail the use of radar systems, communication systems, and RF analyzers for UAV detection and discusses the limitations of these techniques. [72] explores the deployment of software-defined radio (SDR) in unmanned aerial vehicle (UAV) applications, focusing on the areas of communication, security, detection, classification, and localization. The article discusses the application of SDR technology in UAS and provides recent advances and challenges in related technologies. The detection and recognition performance of UAV based on radar technology is shown in Table 8.

The great localization and detection precision of radar detection makes it superior to other techniques. Radar can detect autonomous unmanned aerial vehicles (UAVs) and UAVs with noise suppression capabilities, unlike acoustic or radio frequency detection. Rain, fog, and dust are examples of visual conditions that often have no effect

Table 8

Comparison and summary of UAV detection and recognition technology based on radar technology.

Ref.	Detection target	Method	Performance
[65]	Drone detection and tracking	C-band dual receiver radar system	Well
[66]	Drone detection	35 GHz FMCW drone detection system	Well
[67]	Drone identification	MIMO OFDM radar system	Well
[68]	Drone detection	LFMCW radar system	Precision: 88.82%, Accuracy: 71.43%
[69]	Drone detection	8.75 GHz FMCW drone detection system	Well
[70]	Drone detection	FWCW radar system	Well

**Fig. 11.** Radar-based UAV detection and identification system.

on detection accuracy. Therefore, radar is frequently regarded as the finest option when great precision and dependability are required. Nevertheless, UAVs and other smaller objects cannot be detected by conventional radars since they are built to detect airplanes with a large radar cross-section area (RCS). Secondly, radars might not be able to discern between a stationary reflecting object and a hovering (not moving) or slowly moving UAV. Third, small drone echoes are weak, which may cause false positives or missed positives. Objects such as birds have Doppler signatures similar to drones, and radar often fails drone and bird signals. It is also poor at detecting drones that fly in irregular groups at different speeds. Fourth, radar deployment costs are high. Radar also uses high frequencies that require regulatory clearance, making radar unsuitable for drone detection in many applications.

In addition, in locations where radars already exist, such as airports [73], it may be difficult to install additional radars due to RF interference issues. If part of the spectrum of radar and radio waves overlaps, it will cause signal interference, and the detection performance of radar and the network will be reduced. Some studies have investigated interference between military or government, private radars and 5G wireless access networks [74–76].

5.4. UAV detection and recognition based on infrared characteristics

At night or in other poor light environments, visible-based drone detection technology is no longer applicable due to insufficient light. Infrared detection technology has strong anti-interference ability, good concealment and strong adaptability, and its price and maintenance costs are relatively cheap, especially suitable for the UAV dynamic detection in the low-altitude area.

The existing infrared detection technology mainly includes single frame detection and multi-frame detection. Single frame detection is to detect the target in one frame of infrared image. Relatively speaking, the algorithm is relatively simple, and it can be divided into local information single frame detection and non-local information single frame detection according to the information point of concern. Local information single frame detection is a method to identify the target by

comparing the difference between the target pixel and the surrounding background pixel. Common techniques include the background method [77], the morphological method [78], the gradient derivative method [79] and the contrast method [80]. Non-local information single frame detection technology can find the target by analyzing the whole image, which can avoid the situation that the real target is ignored because the local image features are too significant. Commonly used techniques include the frequency domain method [81] and the classifier method [82].

At present, object detection technology has developed from single image recognition to continuously moving object detection in a video stream. This multi-frame detection technology is based on single-frame detection. By integrating the target position information and space-time information in successive frames, the objects that may be judged as suspected targets in the single frame algorithm are further screened and confirmed. This process can be divided into two main categories: time domain check class and direct determination class [83].

The core of infrared target tracking technology is to use the position information of the target in the current frame, estimate the position of the target in the next frame or the subsequent frame through the prediction model, and generate the trajectory of the target accordingly. This process involves several key steps, including object detection, location prediction, feature extraction and data correlation. The existing infrared target tracking techniques are mainly divided into generative tracking and discriminative tracking. The detection and recognition performance of UAV based on infrared technology is shown in Table 9.

Ding L et al. improved the network structure of the SSD algorithm for infrared small target detection, proposed the use of deep learning methods to detect infrared UAV targets, and designed an adaptive pipeline filter based on time correlation and motion information to correct the recognition results [84]. Aiming at the shortage of UAV infrared imaging sample data sets, taking a small quadrotor UAV as the main research object, Hua Siqi analyzed the detection mode and imaging principle of the infrared thermal imager, studied the imaging characteristics and statistical characteristics of the UAV target, background and noise combined with theoretical calculation and measured

Table 9

Comparison and summary of UAV detection and recognition technology based on infrared technology.

Ref.	Detection target	Method	Performance
[84]	Drone detection and tracking	SSD (Single Shot Multibox Detector)	Recall: 90.7%, Precision: 91.4%
[85]	Drone detection and tracking	3ViBe, Kalman Filter, ANN	Success Rate: 96.1%
[86]	Drone detection and tracking	YOLOv7, StrongSORT	Accuracy: 95.4%, Recall: 87.3%, mAP: 96.1%
[87]	Drone detection and tracking	LCM, RLCM, SORT	mAP: 99.36%

image data, and designed the technical scheme of an infrared vision detection and tracking system suitable for low-altitude airspace [85]. In order to solve the problem that irregular motion (acceleration, sudden stop, etc.) during UAV movement causes a large position deviation of the prediction frame and detection frame, which leads to tracking failure, Liang Xiao proposed an improved StrongSORT tracking algorithm. In the process of boundary matching, BIoU boundary matching algorithm is used to expand its search range. Reduce the impact of motion estimation bias on tracking [86]. For UAV detection under strong noise and complex background interference, Xu Yongli proposed an infrared small target detection algorithm based on CFSFDP (Clustering by Fast Search and Find of Density Peaks) and improved region growth in traditional image analysis. In terms of deep learning, she also proposed an improved YOLOv4 infrared small target detection algorithm [87].

Compared with image recognition technology, infrared technology has the advantages of low cost, strong adaptability to low-visibility environments, and all-weather work. Infrared detection technology has strong anti-interference ability, good concealment and strong adaptability, and its price and maintenance costs are relatively cheap, especially suitable for the UAV dynamic detection in the low-altitude areas. However, in the actual experiment, due to the limitation of its resolution, the accuracy of the measurement using the infrared thermal imager is limited. At the same time, during the flight of an unmanned aircraft, if its own radiation temperature is the same as that of the surrounding environment, the ambient temperature also has a certain impact on infrared measurement.

5.5. Acoustic-based UAV detection and recognition

The rotating blades of drones produce unique acoustic signals that can be used to detect and identify drone models [88]. Propeller blades have a relatively high amplitude and are often used for detection. The process of using microphones to capture acoustic signals and process them to match the drone ID in the database is called fingerprint recognition [89]. Sound Source Localization (SLL) refers to feature extraction after the acquired audio signal is de-noised, filtered, amplified, etc., and the location of the drone is calculated through the corresponding algorithm.

The features extracted from an audio signal mainly include time domain features and frequency domain features. The time domain feature is the most simple and direct feature extraction for the sampled signal, with time as the independent variable. But generally, time domain analysis can only get the power change of the acoustic signal and cannot judge the frequency components contained in the acoustic signal, so it should be analyzed based on the frequency domain characteristics. The time domain signal of the acoustic signal can be converted to the frequency domain by Fourier transform to obtain the corresponding phase spectrum and amplitude spectrum, from which the frequency components of the signal can be analyzed to obtain the energy distribution of different frequency components during this period [90]. Thus, the frequency domain features of the sound signal can be extracted. The characteristics of the frequency domain include the Mel frequency cepstrum coefficient, log-amplitude spectrum, log-Mel filter bank energy, perceptual linear prediction, Gammatone frequency cepstrum coefficient, power regularized cepstrum coefficient, etc.

Mel frequency cepstrum coefficient (MFCC) and linear predictive coding (LPCC) are two common sound signal feature extraction techniques. MFCC is a speech analysis method based on human hearing

mechanisms that analyzes the spectrum of speech by simulating the perception characteristics of human ears to sound signals. The discrete cosine transform (DCT) is usually used to reduce redundant information in the signal, because it retains the main energy of the signal [91], and the flow chart of using MFCC to realize UAV voice detection is shown in the Fig. 12. A Hamming window is usually used in the windowing part of the addition, which is modeled as a signal:

$$Y(n) = X(n) \cdot W(n) \quad (1)$$

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \leq n \leq N-1 \quad (2)$$

where $w(n)$ is the window, n represents the number of samples per frame, $Y(n)$ represents the output signal, $X(n)$ represents the input signal, and $W(n)$ represents the Hamming window.

Frequencies in Hertz are converted to the Mel frequency scale:

$$mel(f) = 2595 \log_{10}(f/700 + 1) \quad (3)$$

where mel denotes the frequency on the Mel scale and f denotes the frequency in Hertz.

The Fast Fourier Transform (FFT) converts each frame of N samples from the time domain to the frequency domain with the following mathematical expression:

$$Y(w) = FFT(h(t) \cdot x(t)) = H(w) \cdot X(w) \quad (4)$$

where $X(w)$ and $H(w)$ are the results after $x(t)$ and $h(t)$ Fourier transforms, respectively.

Mel spectral model:

$$c(n) = \sum_{m=0}^{M-1} \left(\log D(m) \cdot \cos\left(\frac{\pi n(k-0.5)}{M}\right) \right), \quad n = 0, 1, \dots, C-1; \quad 0 \leq k \leq M-1 \quad (5)$$

where $c(n)$ denotes the MFCC coefficients, C is the number of MFCC coefficients, $D(m)$ is the Mayer spectrum of the magnitude spectrum, m is the m th triangular filter coefficient, k is the sample metric, and M is the total number of samples.

In LPCC, the cepstrum coefficients are calculated mainly by linear prediction (LP) and filter banks. The specific steps of calculating the cepstrum coefficient through linear prediction include pre-emphasis, frame division, window addition, etc. The flow chart is shown in Fig. 13.

Among them, pre-weighting is to perform high-pass filtering on the received signal to improve the high-frequency part and reduce the impact of the low-frequency part on the subsequent process. Next, the pre-processed signal is divided into several frames by frame blocking technology, and there is a certain overlap between frames to prepare for subsequent processing. The main purpose of adding Windows is to eliminate edge effects. In this process, the Hamming window is usually used to achieve this purpose. The hamming window can smoothly weight the signal in the time domain to suppress the endpoint leakage of the signal [92]. Through the autocorrelation analysis of each frame of the windowed signal, we can find the repeated pattern in the signal or identify the fundamental frequency hidden in the harmonic frequency of the signal, which is helpful to detect the periodic signal components confused in the random signal. LPC analysis is to fit the spectrum envelope of a speech signal by linear prediction coefficient, and finally get the LPCC coefficient of each frame speech signal by cepstrum conversion.

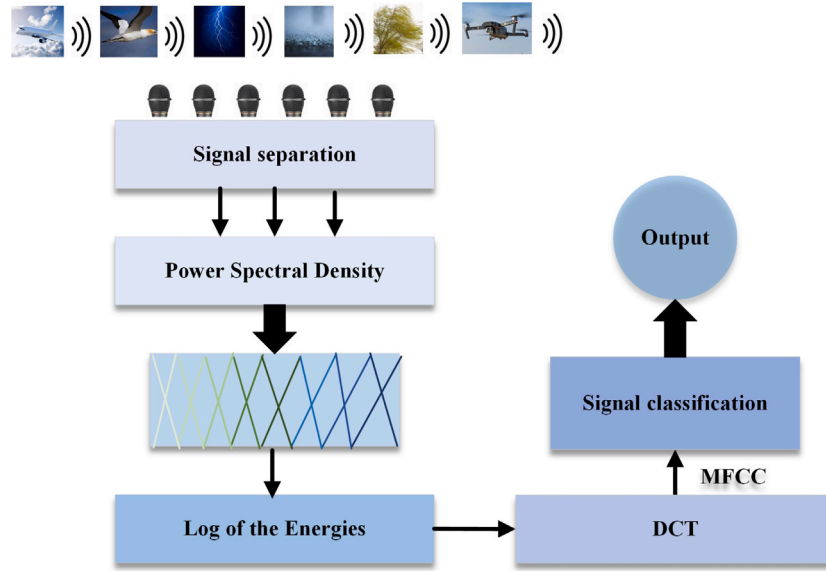


Fig. 12. Flow chart of UAV voice detection using MFCC.

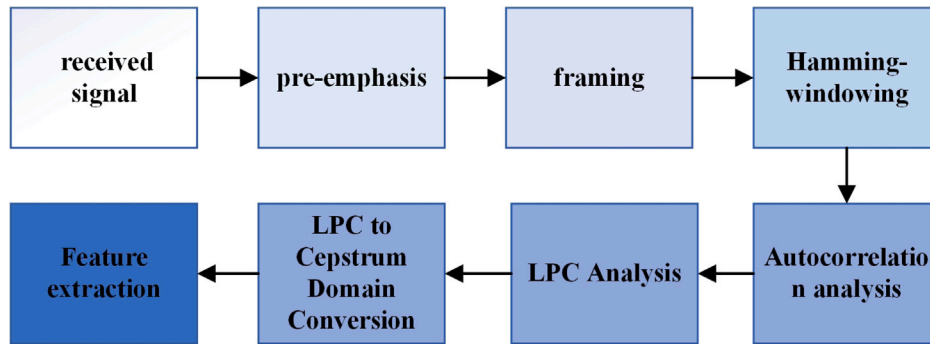


Fig. 13. LPCC algorithm flow chart.

Acoustic-based UAV detection schemes mainly use one or more microphones for acoustic signal recording; [93] use one microphone to collect UAV acoustic signals; [94] use four microphones; and [95] use 120 microphones. Individual microphones do not detect well, so strategically placed arrays of microphones are often used to capture sound from all directions.

To capture the location of the detected drone, array signal processing algorithms, such as MUSIC (Multiple Signal Classification) and beamforming, can be used to estimate the direction of arrival (DOA) of the acoustic signal. In a joint experiment with the German army and police, Frank Christnacher et al. proposed an acoustic system consisting of an array of microphones that could detect and locate approximate positions [94]. Aiming at detecting and locating drones in non-line-of-sight (NLoS) scenarios, Yimiao Sun et al. [96] proposed AIM (Acoustic Inertial Measurement), an indoor UAV positioning and tracking technology based entirely on acoustic signals. The core idea is to use the acoustic characteristics generated by the drone itself to locate and estimate the trajectory of motion. In a complex 10 m×10 m indoor scene, the average positioning error of AIM is 1.89 m, 46% lower than that of commercial UWB systems. Al-Emadi et al. proposed a UAV detection and recognition method using deep learning technologies, such as convolutional neural networks (CNN), recurrent neural networks (RNN) and convolutional recurrent neural networks (CRNN), and used the unique acoustic fingerprint detection of UAVs to identify UAVs [97]. Sheng Wu et al. [98] proposed two improved DOA (Direction of Arrival) estimation algorithms to solve the performance

degradation problem of traditional algorithms in low signal-to-noise ratio (SNR) environments, and designed a DOA algorithm based on time delay estimation neural network, which broke through the hardware limitations of traditional TDOA algorithms and the performance bottleneck under low SNR conditions. [99] developed an acoustic system for detecting, tracking and classifying UAV based on propeller noise, named DADS. The system consists of three or more microphones that transmit data in real time to a central computer for data processing via WIFI. Wang Wenshuai et al. proposed a UAV recognition method based on the FastICA algorithm to solve the problem of UAV sound recognition in a strong interference environment. By using the FastICA algorithm to conduct independent component analysis of sound signals, the UAV sound features can be effectively extracted, so as to achieve accurate recognition of UAV sound [100]. Zahoor Uddin et al. proposed an effective independent component analysis (ICA) unsupervised machine learning method for detecting various sounds in real scenes, including birds, aircraft, thunderstorms, rain, wind, and drones. After unmixing the signals, ICA was used to extract features such as Mel frequency inverse spectral coefficient (MFCC), power spectral density (PSD), and root mean square value (RMS) of PSD, and the detected signals were classified by support vector machine (SVM) and K-nearest neighbor (KNN) to determine whether there were UAV signals in the detected signals [101]. After that, Zahoor Uddin et al. used an acoustic signal processing method based on independent vector analysis (IVA) to detect active electromagnetic interference in the field. This method can simultaneously detect multiple amateur drones in the sensing field

Table 10

Comparison and summary of acoustic-based UAV detection and recognition techniques.

Ref.	Detection target	Method	Performance
[94]	Drone detection	MUSIC, DOA	Well
[96]	Drone detection and tracking	AIM	Average positioning error: 1.89 m
[97]	Drone detection and identification	DNN, GAN	Precision: 98.71%, Recall: 96.44%, F1 score: 97.53%, Accuracy: 95.63%
[98]	Drone tracking	Improved DOA algorithms	Well
[99]	Drone detection	SRP-PHAT	Precision: 3.2 (deg), DOA finding range: 360 m
[100]	Drone detection	FASTICA	Accuracy: 90%–100%
[101]	Drone detection	SVM, LPCC, MFCC	Accuracy: 96.7%
[102]	Drone detection	IVA, KNN, MFCC, PSD, SVM	Accuracy: 99.5%

in the presence of strong interference sources, which is more efficient than independent component analysis (ICA) [102]. The acoustic-based UAV detection and recognition performance is shown in Table 10.

The use of acoustic signals to detect drones is relatively inexpensive, unaffected in low light and various weather conditions, does not require targeting drones, and is suitable for remote areas without background noise. The small wing of the UAV is small, the sound energy generated is low, and the noise and other interfering factors in the environment make its signal decay faster in the environment, and the signal-to-noise ratio is low. This directly affects the detection and positioning distance of the system and limits the effective working range of the UAV. When there is a complex environmental noise background, the acoustic-based UAV detection accuracy will also be seriously reduced. UAV audio signal recognition still faces many challenges, including but not limited to the complexity of environmental noise, the mismatch between training and recognition environments, and the robustness of the model. Future research needs to further explore more effective noise reduction techniques and model optimization strategies to improve the accuracy of speech recognition in complex noise environments.

5.6. Multi-modal fusion detection UAV method

Unmanned Aerial Vehicle (UAV) detection methods that rely on a single modality, such as image processing, RF signal detection, or radar technology, offer significant advantages in specific scenarios but often face limitations in practical applications. For instance, image processing-based detection methods can suffer from severe performance degradation in low-light conditions, haze, or when occlusions occur. RF signal detection is prone to interference or blockage, particularly in urban environments. Similarly, radar technology can encounter challenges in complex terrains (e.g., areas with trees or tall buildings) due to reflection interference, which complicates target identification. Acoustic detection methods, on the other hand, are often unstable in noisy environments. To address the limitations of single-modality methods in terms of detection range and classification accuracy, multimodal fusion technology integrates complementary information from different modalities, compensating for missing data by leveraging the perspectives provided by other sensors [103]. This approach is akin to a person using multiple senses: one eye observes the shape and size of an object, the other eye provides depth perception, and the nose detects its scent. Similarly, multimodal fusion offers a more robust and efficient solution for UAV detection. It not only improves detection accuracy but also significantly reduces false positive and false negative rates, particularly in complex environments.

Multimodal fusion detection is designed to process and correlate information from multiple sensors from a holistic perspective [104]. The flow chart of multi-mode fusion detection is shown in Fig. 14. For example, in 2022, Fredrik Svanström [105] designed a real-time multi-sensor UAV detection system that incorporated video cameras, thermal imaging cameras, fisheye cameras, microphones, and ADS-B receivers (for receiving aircraft broadcast signals), among other sensors. A new dataset, including birds, drones, aircraft, and helicopters, was introduced, along with the sensor-to-target distance, to analyze classification performance as a function of distance. Due to the limited computing resources of small UAVs and the restricted field of view of

sensors, it is critical to achieve efficient and real-time target detection and tracking. In 2023, Chuanbeibei Shi et al. [106] proposed an early fusion strategy based on LiDAR point cloud and RGB images. Their method processed point cloud data into a three-channel tensor (XYZ) and combined it with RGB images to form a six-channel tensor, which was then fed into a deep neural network. Using the Gaussian YOLOv3 network, they achieved 94.6% mAP on the test set for UAV target detection. Experiments under different lighting conditions, such as low light and complete darkness, demonstrated the robustness of this multimodal fusion method in challenging environments. To further enhance UAV classification accuracy, Nikos Sakellariou et al. [107] proposed a deep neural network in 2024 based on multimodal sensor data fusion. Their approach integrated feature maps from thermal imaging, visible light sensors, and radar data to classify aerial objects as either “drones” or “non-drones” (e.g., birds, aircraft). Addressing the synchronization challenges often faced in multimodal UAV detection systems, Shenghai Yuan et al. [18] introduced MMAUD in 2024, a comprehensive multimodal anti-UAV dataset. They proposed a multimodal fusion method combining LiDAR, RGB cameras, GPS, and IMU data to improve detection accuracy and real-time performance. Their method achieved a detection accuracy of 94.6% mAP in real-world scenarios, approximately 5% higher than single-sensor networks, albeit with a tenfold increase in preprocessing time to meet real-time detection requirements. Ildi Alla et al. [108], in WiSec 2024, proposed a novel neural sensor fusion framework that combines audio and infrared video data. Their framework employs Dynamic Weighted Average Fusion (DWAF) and Multi-Modal Tree Fusion (MMTF) methods to significantly enhance the accuracy and robustness of UAV detection. In the data fusion process, the DWAF method dynamically adjusts weights based on the confidence levels of the audio and video classification models. Meanwhile, the MMTF method, which utilizes a Random Forest classifier to integrate audio and video predictions, improves detection performance when multimodal data is partially missing. The system achieved rapid predictions (only 0.382 s) while maintaining high accuracy in complex environments, such as those with impaired audio and video data. Notably, the MMTF method achieved a detection accuracy of 96.02%.

6. Conclusion

The rapid proliferation of unmanned aerial vehicles (UAVs) has introduced both opportunities and challenges, particularly in areas such as security, privacy, and airspace management. To address the need for effective UAV detection and identification, numerous detection methods have been proposed and extensively studied. This review provides an extensive overview of the current anti-drone detection and identification technologies, encompassing six primary approaches: (1) drone detection based on image processing, (2) drone detection based on RF signals, (3) drone detection based on radar technology, (4) infrared feature-based drone detection, (5) acoustic-based drone detection, and (6) UAV detection based on multimodal fusion.

Despite the significant progress achieved in UAV detection technologies, several critical challenges remain to be addressed. Current systems often face limitations such as performance degradation under adverse environmental conditions, difficulties in detecting small or

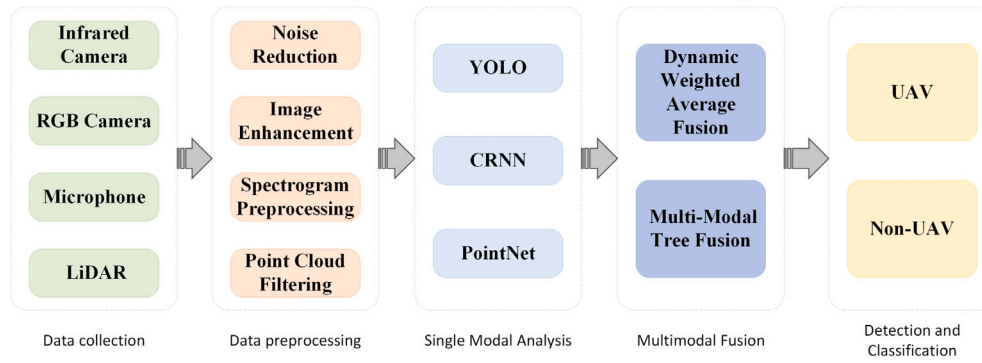


Fig. 14. Flow chart of multi-mode fusion detection UAV method.

stealthy targets, and a lack of large, diverse datasets to support robust model training. Furthermore, the integration of multimodal data from different sensors remains a complex yet promising area of research that requires further exploration. While substantial progress has been made, several critical challenges remain:

1. **Environmental Constraints:** Detection systems often struggle with low visibility, complex backgrounds, and small or stealthy targets.
2. **Technical Limitations:** Sensor performance, limited dataset diversity, and lack of multimodal synchronization hinder robust detection.
3. **Data-Related Issues:** Existing datasets are insufficiently diverse and lack synchronization, limiting model generalization.
4. **Additional Concerns:** Evasion techniques, real-time processing requirements, power consumption, and privacy issues further complicate detection.

Future research directions should focus on the following key aspects:

Data enhancement and diversity

1. **Enhancing Data Collection:** Improve data collection methods and expand dataset diversity to support the development of deep learning models with superior generalization capabilities.
2. **Multimodal Synchronization:** Ensure multimodal synchronization and incorporate large-scale, diverse datasets to facilitate advanced detection techniques.

Environmental adaptation and robustness

1. **Robust Detection Methods:** Develop robust and adaptive detection methods capable of accurate UAV detection under diverse and challenging environmental conditions.
2. **Adverse Conditions Enhancement:** Utilize advanced image processing, multi-spectral imaging, and adaptive filtering to enhance performance in adverse conditions.

Distance-aware and multimodal fusion

1. **Distance-Aware Metrics:** Incorporate distance-aware metrics to improve detection system performance across varying ranges.
2. **Multimodal Fusion Techniques:** Advance multimodal fusion techniques to leverage the complementary strengths of sensors such as radar, infrared, and acoustic systems.

Technical and algorithmic innovations

1. **Efficient Algorithms:** Implement efficient algorithms, hardware acceleration, and robust synchronization mechanisms to enhance system performance.
2. **Privacy and Accuracy:** Develop adaptive detection systems and ensure data anonymization to address privacy concerns while maintaining detection accuracy.

By addressing these challenges, the reliability, accuracy, and scalability of drone detection systems can be significantly improved. Ultimately, these advancements will ensure the protection of people, assets, and critical infrastructure from potential threats posed by the unauthorized use of civilian drones.

CRediT authorship contribution statement

Lan Xu: Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Zhongqiang Luo:** Writing – review & editing, Validation, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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